Tutorial 8: Classical Methods of Optimization

Solutions

Exercise 1: Cost Minimization

We consider the cost function

$$f(x, y) = x^2 + y^2 - 10x - 12y + 151.$$

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$$f(x, y) = x^2 + y^2 - 10x - 12y + 151.$$

The company wants to minimize its cost with respect to x and y.

Exercise 1: Question 1

The first-order conditions are

$$\frac{\partial f}{\partial x} = 2x - 10 = 0, \qquad \frac{\partial f}{\partial y} = 2y - 12 = 0.$$

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Thus, we solve the system

$$\begin{cases} 2x - 10 = 0, \\ 2y - 12 = 0. \end{cases}$$

Hence,

$$\begin{cases} 2x = 10, \\ 2y = 12, \end{cases} \qquad \Longrightarrow \qquad \begin{cases} x = 5, \\ y = 6. \end{cases}$$

Exercise 1: Question 1

So the only critical point is

$$\boxed{(x, y) = (5, 6)}.$$

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Now complete the square:

$$f(x, y) = x^2 - 10x + y^2 - 12y + 151.$$

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So the only critical point is

$$\boxed{(x, y) = (5, 6)}.$$

Now complete the square:

$$f(x, y) = x^2 - 10x + y^2 - 12y + 151.$$

$$f(x, y) = (x - 5)^2 - 25 + (y - 6)^2 - 36 + 151.$$

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So the only critical point is

$$\boxed{(x, y) = (5, 6)}.$$

Now complete the square:

$$f(x, y) = x^2 - 10x + y^2 - 12y + 151.$$

$$f(x, y) = (x - 5)^2 - 25 + (y - 6)^2 - 36 + 151.$$

Therefore,

$$f(x, y) = (x - 5)^2 + (y - 6)^2 + 90.$$

Exercise 1: Question 1

Since

$$(x-5)^2 \geq 0, \qquad (y-6)^2 \geq 0,$$

we have

$$f(x,y) \geq 90.$$

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Since

$$(x-5)^2 \geq 0, \qquad (y-6)^2 \geq 0,$$

we have

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The equality holds when

$$x-5=0, \qquad y-6=0.$$

Since

$$(x-5)^2 \geq 0, \qquad (y-6)^2 \geq 0,$$

we have

$$f(x, y) \geq 90.$$

The equality holds when

$$x - 5 = 0, \qquad y - 6 = 0.$$

Thus,

$$\boxed{x = 5, \qquad y = 6}.$$

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Since

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we have

$$f(x,y) \geq 90.$$

The equality holds when

$$x-5=0, \qquad y-6=0.$$

Thus,

$$\boxed{x=5, \qquad y=6}.$$

The minimum cost is

$$\boxed{f_{\min}=90}.$$

Exercise 1: Question 2

The Hessian matrix is

$$H_f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

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The leading principal minors are

$$\Delta_1 = 2 > 0,$$

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The Hessian matrix is

$$H_f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The leading principal minors are

$$\Delta_1 = 2 > 0,$$

and

$$\Delta_2 = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0.$$

Exercise 1: Question 2

Since

$$\Delta_1 > 0, \qquad \Delta_2 > 0,$$

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Since

$$\Delta_1 > 0, \qquad \Delta_2 > 0,$$

we conclude that

$$H_f(x, y)$$

is positive definite.

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Since

$$\Delta_1 > 0, \qquad \Delta_2 > 0,$$

we conclude that

$$H_f(x, y)$$

is positive definite. Therefore, the critical point is a minimum. Hence,

$$\boxed{(x, y) = (5, 6)}$$

is indeed the cost-minimizing point.

Exercise 2: Profit Maximization

We are given

$$C(q, l) = 2l^2 + 5q^2 - 20q + 400,$$

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$$C(q, l) = 2l^2 + 5q^2 - 20q + 400,$$

and

$$p = 100 - 3q + 4l.$$

Exercise 2: Profit Maximization

We are given

$$C(q, I) = 2I^2 + 5q^2 - 20q + 400,$$

and

$$p = 100 - 3q + 4I.$$

Here q is the quantity produced, and I is the quality index.

Exercise 2: Question 1

Total revenue is

$$R = pq.$$

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Since

$$p = 100 - 3q + 4I,$$

we get

$$R(q, I) = (100 - 3q + 4I)q.$$

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Total revenue is

$$R = pq.$$

Since

$$p = 100 - 3q + 4I,$$

we get

$$R(q, I) = (100 - 3q + 4I)q.$$

Therefore,

$$\boxed{R(q, I) = 100q - 3q^2 + 4Iq}.$$

Exercise 2: Question 2

Profit is

$$\pi = R - C.$$

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Thus,

$$\pi(q, I) = 100q - 3q^2 + 4Iq - \left(2I^2 + 5q^2 - 20q + 400\right).$$

Exercise 2: Question 2

Profit is

$$\pi = R - C.$$

Thus,

$$\pi(q, I) = 100q - 3q^2 + 4Iq - \left(2I^2 + 5q^2 - 20q + 400\right).$$

Hence,

$$\boxed{\pi(q, I) = 120q - 8q^2 + 4Iq - 2I^2 - 400}.$$

Exercise 2: Question 2

The first-order conditions are

$$\frac{\partial \pi}{\partial q} = 120 - 16q + 4l = 0, \qquad \frac{\partial \pi}{\partial l} = 4q - 4l = 0.$$

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$$\frac{\partial \pi}{\partial q} = 120 - 16q + 4l = 0, \qquad \frac{\partial \pi}{\partial l} = 4q - 4l = 0.$$

Thus, we solve

$$\begin{cases} 120 - 16q + 4l = 0, \\ 4q - 4l = 0. \end{cases}$$

Exercise 2: Question 2

The first-order conditions are

$$\frac{\partial \pi}{\partial q} = 120 - 16q + 4l = 0, \qquad \frac{\partial \pi}{\partial l} = 4q - 4l = 0.$$

Thus, we solve

$$\begin{cases} 120 - 16q + 4l = 0, \\ 4q - 4l = 0. \end{cases}$$

From the second equation,

$$4q - 4l = 0 \qquad \Longrightarrow \qquad l = q.$$

Exercise 2: Question 2

Substitute $l = q$ into the first equation:

$$120 - 16q + 4q = 0.$$

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$$120 - 16q + 4q = 0.$$

Thus,

$$120 - 12q = 0.$$

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Substitute $l = q$ into the first equation:

$$120 - 16q + 4q = 0.$$

Thus,

$$120 - 12q = 0.$$

Hence,

$$q = 10.$$

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Substitute $l = q$ into the first equation:

$$120 - 16q + 4q = 0.$$

Thus,

$$120 - 12q = 0.$$

Hence,

$$q = 10.$$

Since $l = q$, we obtain

$$l = 10.$$

Exercise 2: Question 2

Substitute $l = q$ into the first equation:

$$120 - 16q + 4q = 0.$$

Thus,

$$120 - 12q = 0.$$

Hence,

$$q = 10.$$

Since $l = q$, we obtain

$$l = 10.$$

So the only critical point is

$$\boxed{(q, l) = (10, 10)}.$$

Exercise 2: Question 2

Now write the profit function as

$$\pi(q, l) = 200 - 6(q - 10)^2 - 2(q - l)^2.$$

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Now write the profit function as

$$\pi(q, l) = 200 - 6(q - 10)^2 - 2(q - l)^2.$$

Since

$$(q - 10)^2 \geq 0, \qquad (q - l)^2 \geq 0,$$

we have

$$-6(q - 10)^2 \leq 0, \qquad -2(q - l)^2 \leq 0.$$

Exercise 2: Question 2

Now write the profit function as

$$\pi(q, I) = 200 - 6(q - 10)^2 - 2(q - I)^2.$$

Since

$$(q - 10)^2 \geq 0, \qquad (q - I)^2 \geq 0,$$

we have

$$-6(q - 10)^2 \leq 0, \qquad -2(q - I)^2 \leq 0.$$

Therefore,

$$\pi(q, I) \leq 200.$$

Exercise 2: Question 2

The equality holds when

$$q - 10 = 0, \qquad q - l = 0.$$

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The equality holds when

$$q - 10 = 0, \qquad q - l = 0.$$

Therefore,

$$\boxed{q = 10, \qquad l = 10}.$$

The maximum profit is

$$\boxed{\pi_{\max} = 200}.$$

Exercise 2: Question 3

The Hessian matrix is

$$H_\pi(q, l) = \begin{pmatrix} -16 & 4 \\ 4 & -4 \end{pmatrix}.$$

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The leading principal minors are

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Exercise 2: Question 3

The Hessian matrix is

$$H_\pi(q, l) = \begin{pmatrix} -16 & 4 \\ 4 & -4 \end{pmatrix}.$$

The leading principal minors are

$$\Delta_1 = -16 < 0,$$

and

$$\Delta_2 = \begin{vmatrix} -16 & 4 \\ 4 & -4 \end{vmatrix} = 64 - 16 = 48 > 0.$$

Exercise 2: Question 3

Since

$$\Delta_1 < 0, \qquad \Delta_2 > 0,$$

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Since

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we conclude that

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is negative definite.

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Since

$$\Delta_1 < 0, \qquad \Delta_2 > 0,$$

we conclude that

$$H_\pi(q, l)$$

is negative definite. Therefore, the local extremum is a global maximum. Hence,

$$\boxed{q = 10, \qquad l = 10, \qquad \pi_{\text{max}} = 200}.$$

Exercise 3: Constrained Optimization

We consider

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

Exercise 3: Constrained Optimization

We consider

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

We solve it using two methods.

Method 1: Direct Substitution

From the constraint,

$$x + y + 2z = 10,$$

we get

$$y = 10 - x - 2z.$$

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From the constraint,

$$x + y + 2z = 10,$$

we get

$$y = 10 - x - 2z.$$

Substitute into f:

$$F(x, z) = f(x, 10 - x - 2z, z).$$

Method 1: Direct Substitution

From the constraint,

$$x + y + 2z = 10,$$

we get

$$y = 10 - x - 2z.$$

Substitute into f:

$$F(x, z) = f(x, 10 - x - 2z, z).$$

Since

$$f(x, y, z) = (x - y)^2 + 5z^2,$$

we obtain

$$F(x, z) = (x - (10 - x - 2z))^2 + 5z^2.$$

Method 1: Direct Substitution

From the constraint,

$$x + y + 2z = 10,$$

we get

$$y = 10 - x - 2z.$$

Substitute into f:

$$F(x, z) = f(x, 10 - x - 2z, z).$$

Since

$$f(x, y, z) = (x - y)^2 + 5z^2,$$

we obtain

$$F(x, z) = (x - (10 - x - 2z))^2 + 5z^2.$$

Thus,

$$F(x, z) = 4(x + z - 5)^2 + 5z^2.$$

Method 1: Critical Point

We compute

$$F_x = 8(x + z - 5),$$

and

$$F_z = 8(x + z - 5) + 10z.$$

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The critical point satisfies

$$\begin{cases} 8(x + z - 5) = 0, \\ 8(x + z - 5) + 10z = 0. \end{cases}$$

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From the first equation,

$$x + z - 5 = 0.$$

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From the first equation,

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Substituting into the second equation gives

$$10z = 0,$$

hence

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The critical point satisfies

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From the first equation,

$$x + z - 5 = 0.$$

Substituting into the second equation gives

$$10z = 0,$$

hence

$$z = 0.$$

Therefore,

$$x = 5.$$

Method 1: Candidate

Using

$$y = 10 - x - 2z,$$

we get

$$y = 10 - 5 - 0 = 5.$$

Method 1: Candidate

Using

$$y = 10 - x - 2z,$$

we get

$$y = 10 - 5 - 0 = 5.$$

Hence, the constrained critical point is

$$\boxed{(x, y, z) = (5, 5, 0)}.$$

Method 1: Second-Order Test

The Hessian matrix of F is

$$H_F(x,z) = \begin{pmatrix} F_{xx} & F_{xz} \\ F_{zx} & F_{zz} \end{pmatrix}.$$

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$$H_F(x, z) = \begin{pmatrix} F_{xx} & F_{xz} \\ F_{zx} & F_{zz} \end{pmatrix}.$$

Since

$$F(x, z) = 4(x + z - 5)^2 + 5z^2,$$

we get

$$F_{xx} = 8, \qquad F_{xz} = 8, \qquad F_{zz} = 18.$$

Method 1: Second-Order Test

The Hessian matrix of F is

$$H_F(x,z) = \begin{pmatrix} F_{xx} & F_{xz} \\ F_{zx} & F_{zz} \end{pmatrix}.$$

Since

$$F(x,z) = 4(x+z-5)^2 + 5z^2,$$

we get

$$F_{xx} = 8, \qquad F_{xz} = 8, \qquad F_{zz} = 18.$$

Thus,

$$H_F(x,z) = \begin{pmatrix} 8 & 8 \\ 8 & 18 \end{pmatrix}.$$

Method 1: Nature of the Extremum

The leading principal minors are

$$\Delta_1 = 8 > 0,$$

and

$$\Delta_2 = \begin{vmatrix} 8 & 8 \\ 8 & 18 \end{vmatrix} = 144 - 64 = 80 > 0.$$

Method 1: Nature of the Extremum

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Therefore, $H_F(x, z)$ is positive definite.

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Therefore, $H_F(x, z)$ is positive definite. Hence,

$$(x, z) = (5, 0)$$

is a strict local (and global) minimum of F.

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Therefore, $H_F(x, z)$ is positive definite. Hence,

$$(x, z) = (5, 0)$$

is a strict local (and global) minimum of F. Therefore,

$$\boxed{(x, y, z) = (5, 5, 0)}$$

is a global constrained minimum of f.

Method 1: Minimum Value

At the minimum point,

$$f(5,5,0) = 5^2 - 2(5)(5) + 5^2 + 5(0)^2.$$

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$$f(5,5,0) = 5^2 - 2(5)(5) + 5^2 + 5(0)^2.$$

Thus,

$$f(5,5,0) = 25 - 50 + 25 = 0.$$

Method 1: Minimum Value

At the minimum point,

$$f(5,5,0) = 5^2 - 2(5)(5) + 5^2 + 5(0)^2.$$

Thus,

$$f(5,5,0) = 25 - 50 + 25 = 0.$$

Therefore,

$$\boxed{f_{\min} = 0}.$$

Method 2: Lagrange Multipliers

We consider

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

Method 2: Lagrange Multipliers

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$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

Let

$$g(x, y, z) = x + y + 2z - 10.$$

Method 2: Lagrange Multipliers

We consider

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

Let

$$g(x, y, z) = x + y + 2z - 10.$$

The Lagrangian is

$$\mathcal{L}(x, y, z, \lambda) = f(x, y, z) - \lambda g(x, y, z).$$

Method 2: Lagrange Multipliers

We consider

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2,$$

subject to

$$x + y + 2z = 10.$$

Let

$$g(x, y, z) = x + y + 2z - 10.$$

The Lagrangian is

$$\mathcal{L}(x, y, z, \lambda) = f(x, y, z) - \lambda g(x, y, z).$$

Thus,

$$\mathcal{L} = x^2 - 2xy + y^2 + 5z^2 - \lambda(x + y + 2z - 10).$$

Method 2: Necessary Conditions

The necessary condition (since $\nabla g \neq 0$) for a constrained extremum is

$$\nabla f = \lambda \nabla g.$$

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Equivalently,

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Method 2: Necessary Conditions

The necessary condition (since $\nabla g \neq 0$) for a constrained extremum is

$$\nabla f = \lambda \nabla g.$$

Equivalently,

$$\begin{cases} \mathcal{L}_x = 0, \\ \mathcal{L}_y = 0, \\ \mathcal{L}_z = 0, \\ \mathcal{L}_\lambda = 0. \end{cases}$$

Therefore,

$$\begin{cases} 2x - 2y - \lambda = 0, \\ -2x + 2y - \lambda = 0, \\ 10z - 2\lambda = 0, \\ x + y + 2z = 10. \end{cases}$$

Method 2: Solving the System

From the first two equations,

$$\lambda = 2x - 2y, \qquad \lambda = -2x + 2y.$$

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From the first two equations,

$$\lambda = 2x - 2y, \qquad \lambda = -2x + 2y.$$

Thus,

$$2x - 2y = -2x + 2y.$$

Hence,

$$4x - 4y = 0,$$

so

$$x = y.$$

Method 2: Solving the System

From the first two equations,

$$\lambda = 2x - 2y, \qquad \lambda = -2x + 2y.$$

Thus,

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so

$$x = y.$$

Then

$$\lambda = 2x - 2y = 0.$$

Method 2: Candidate Point

From

$$10z - 2\lambda = 0$$

and $\lambda = 0$, we get

$$z = 0.$$

Method 2: Candidate Point

From

$$10z - 2\lambda = 0$$

and $\lambda = 0$, we get

$$z = 0.$$

Using the constraint,

$$x + y + 2z = 10,$$

we obtain

$$x + y = 10.$$

Method 2: Candidate Point

From

$$10z - 2\lambda = 0$$

and $\lambda = 0$, we get

$$z = 0.$$

Using the constraint,

$$x + y + 2z = 10,$$

we obtain

$$x + y = 10.$$

Since $x = y$,

$$2x = 10.$$

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$$10z - 2\lambda = 0$$

and $\lambda = 0$, we get

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Using the constraint,

$$x + y + 2z = 10,$$

we obtain

$$x + y = 10.$$

Since $x = y$,

$$2x = 10.$$

Therefore,

$$x = 5, \qquad y = 5, \qquad z = 0.$$

Method 2: Candidate Point

From
$$10z - 2\lambda = 0$$
and $\lambda = 0$, we get
$$z = 0.$$
Using the constraint,
$$x + y + 2z = 10,$$
we obtain
$$x + y = 10.$$
Since $x = y$,
$$2x = 10.$$
Therefore,
$$x = 5, \qquad y = 5, \qquad z = 0.$$
Thus, the constrained critical point is
$$\boxed{(x, y, z) = (5, 5, 0)}.$$

Method 2: Second-Order Test

The Hessian matrix of the Lagrangian with respect to (x, y, z) is

$$H_{\mathcal{L}} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 10 \end{pmatrix}.$$

Method 2: Second-Order Test

The Hessian matrix of the Lagrangian with respect to (x, y, z) is

$$H_{\mathcal{L}} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 10 \end{pmatrix}.$$

Also,

$$\nabla g = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Method 2: Second-Order Test

The Hessian matrix of the Lagrangian with respect to (x, y, z) is

$$H_{\mathcal{L}} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 10 \end{pmatrix}.$$

Also,

$$\nabla g = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

The sufficient condition will be checked on the admissible variations.

Test 1: Determinantal Equation

The determinant-root test is given by

$$\begin{vmatrix} L_{11} - z_i & L_{12} & L_{13} & g_1 \\ L_{21} & L_{22} - z_i & L_{23} & g_2 \\ L_{31} & L_{32} & L_{33} - z_i & g_3 \\ g_1 & g_2 & g_3 & 0 \end{vmatrix} = 0.$$

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Here, the z_i's are the roots of the test, not the variable z.

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The determinant-root test is given by

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Here, the z_i's are the roots of the test, not the variable z. In our case,

$$H_{\mathcal{L}} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 10 \end{pmatrix}, \qquad \nabla g = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Test 1: Determinantal Equation

Therefore,

$$\begin{vmatrix} 2 - z_i & -2 & 0 & 1 \\ -2 & 2 - z_i & 0 & 1 \\ 0 & 0 & 10 - z_i & 2 \\ 1 & 1 & 2 & 0 \end{vmatrix} = 0.$$

Test 1: Determinantal Equation

Therefore,

$$\begin{vmatrix} 2-z_i & -2 & 0 & 1 \\ -2 & 2-z_i & 0 & 1 \\ 0 & 0 & 10-z_i & 2 \\ 1 & 1 & 2 & 0 \end{vmatrix} = 0.$$

Expanding the determinant gives

$$-2(z_i-4)(3z_i-10) = 0.$$

Test 1: Determinantal Equation

Therefore,

$$\begin{vmatrix} 2-z_i & -2 & 0 & 1 \\ -2 & 2-z_i & 0 & 1 \\ 0 & 0 & 10-z_i & 2 \\ 1 & 1 & 2 & 0 \end{vmatrix} = 0.$$

Expanding the determinant gives

$$-2(z_i-4)(3z_i-10)=0.$$

Thus,

$$z_1 = 4, \qquad z_2 = \frac{10}{3}.$$

Test 1: Nature of the Extremum

We have

$$z_1 = 4 > 0, \qquad z_2 = \frac{10}{3} > 0.$$

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Consequently,

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Since

$$\nabla g = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix},$$

we get

$$u + v + 2w = 0.$$

Test 2: Quadratic Form

We compute

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Final Conclusion

We have proved by two second-order tests that

$$\boxed{(5, 5, 0)}$$

is a constrained local (and global) minimum.